Abstract We introduce the Lantern architecture, a sequence model that dispenses with recurrence entirely and relies instead on a gated retrieval mechanism. On three benchmark translation tasks the model matches the accuracy of recurrent baselines while training in less than a third of the wall-clock time. We attribute the speedup to the absence of sequential dependencies along the time axis, which allows every position to be computed in parallel.

1 Introduction Recurrent networks process a sequence one position at a time, carrying a hidden state forward. This sequential structure is the source of both their strength and their central limitation: the computation for position t cannot begin until position t-1 has finished. Memory constraints then limit batching across examples, and long sequences become expensive to train.

The Lantern model removes the recurrent connection altogether. Each position attends over every other position through a learned similarity function, so the path length between any two positions is constant rather than linear in their distance. Constant path length is what makes long-range dependencies learnable in practice.

2 Architecture The encoder is a stack of six identical layers. Each layer has two sub-
layers: a multi-head retrieval block and a position-wise feed-forward network. We employ a residual connection around each sub-layer, followed by layer normalisation. All sub-layers produce outputs of dimension 512.

2.1 Gated retrieval A retrieval function maps a query and a set of key-value pairs to an output. The output is a weighted sum of the values, where the weight assigned to each value is computed by a compatibility function of the query with the corresponding key. We scale the dot products by the inverse square root of the key dimension, which keeps gradients well-conditioned as dimension grows.

2.2 Multiple heads Rather than performing a single retrieval with full-dimensional keys, we project the queries, keys and values h times with different learned projections. Multiple heads allow the model to attend to information from different representation subspaces at different positions, which a single averaged head would blur together.

3 Training We trained on the standard corpus of 4.5 million sentence pairs. Sentences were encoded with a shared vocabulary of 37000 tokens. Each training batch contained approxi-
mately 25000 source tokens and 25000 target tokens. We used the Adam optimiser with a learning rate that increases linearly for the first 4000 steps and
decays thereafter.

4 Results On the English-to-German task the Lantern model reaches 28.4 BLEU, improving over the best previously reported result by more than 2 BLEU. On English-to-French it reaches 41.8 BLEU after training for 3.5 days on eight GPUs, a small fraction of the training cost of the recurrent models it is compared against.

5 Conclusion We presented the Lantern architecture, the first sequence model based entirely on gated retrieval. The model trains substantially faster than architectures based on recurrent layers, and it sets a new state of the art on two translation tasks. We are excited about applying it to inputs other than text.